

SOC Lab: Attack Simulation and Network Traffic Analysis

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Tools Used: Kali Linux, Ubuntu, Nmap, Nikto, Wireshark

1. Introduction

This project simulates a basic Security Operations Center (SOC) environment to demonstrate how attackers perform reconnaissance and how defenders detect malicious activity. A virtual lab was created using Kali Linux as the attacker and Ubuntu as the target system.

The objective of this project is to identify open services, detect vulnerabilities, and analyze network traffic generated during simulated attacks.

2. Lab Setup

The lab environment consists of two virtual machines running on UTM:

- **Attacker Machine:** Kali Linux (192.168.64.4)
- **Victim Machine:** Ubuntu Server (192.168.64.6)

Both machines are connected through a shared virtual network, allowing communication between them.

The victim machine was configured with an Apache web server to simulate a real-world target system.

3. Attack Simulation

3.1 Port Scanning (Nmap)

A TCP SYN scan was performed using Nmap to identify open ports on the target system.

Command used:

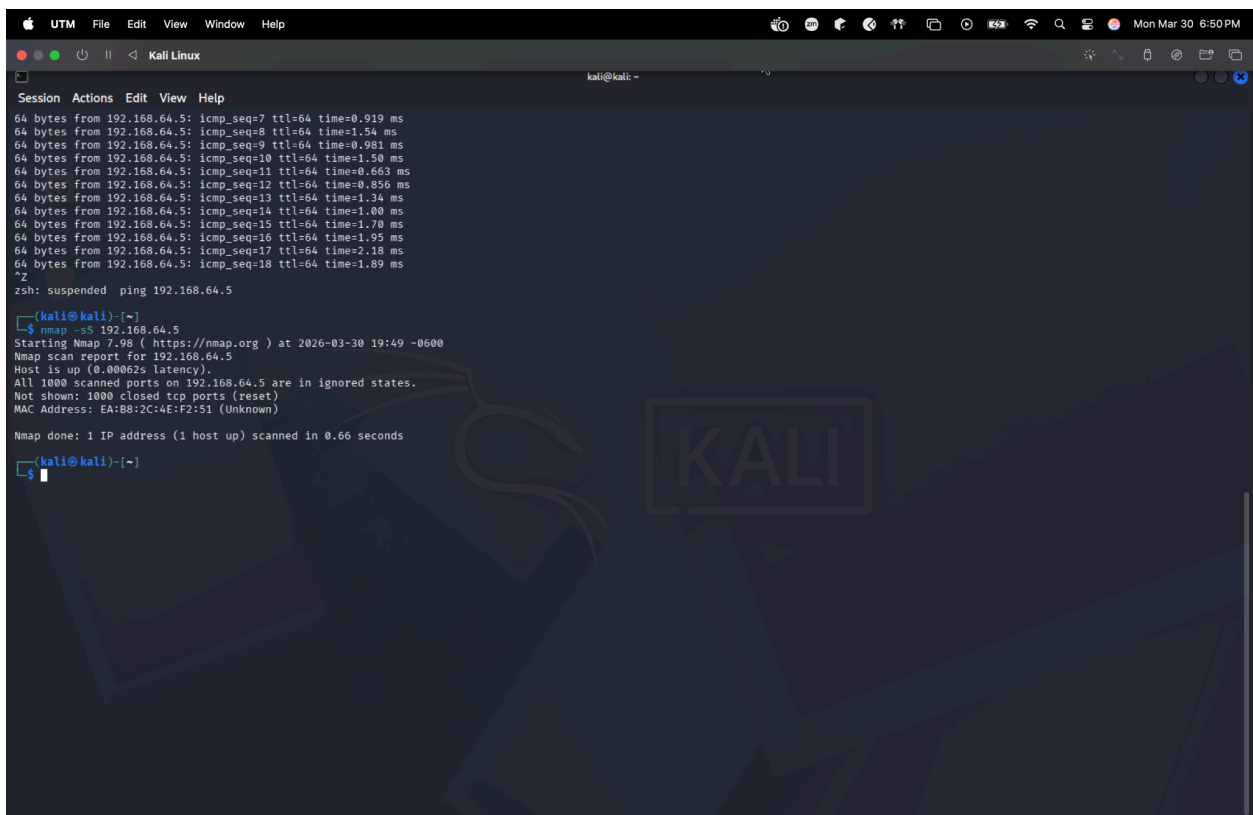
```
nmap -sS 192.168.64.6
```

Initially, all ports on the target system were closed. After installing the Apache web server, port **80 (HTTP)** was detected as open.

This demonstrates how attackers discover exposed services on a system.

Screenshot(s):

nmap before:



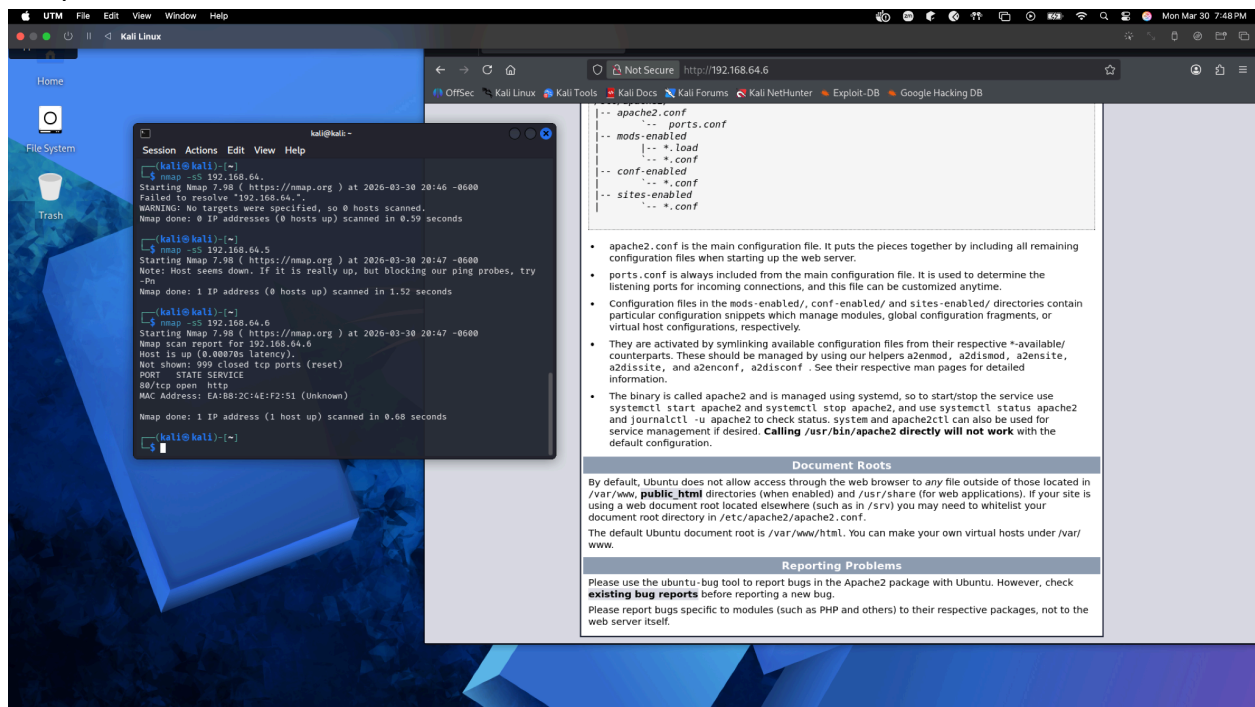
```
UTM File Edit View Window Help
Kali Linux
kali@kali: ~
Session Actions Edit View Help
64 bytes from 192.168.64.5: icmp_seq=7 ttl=64 time=0.919 ms
64 bytes from 192.168.64.5: icmp_seq=8 ttl=64 time=1.54 ms
64 bytes from 192.168.64.5: icmp_seq=9 ttl=64 time=0.981 ms
64 bytes from 192.168.64.5: icmp_seq=10 ttl=64 time=1.50 ms
64 bytes from 192.168.64.5: icmp_seq=11 ttl=64 time=0.663 ms
64 bytes from 192.168.64.5: icmp_seq=12 ttl=64 time=0.856 ms
64 bytes from 192.168.64.5: icmp_seq=13 ttl=64 time=1.34 ms
64 bytes from 192.168.64.5: icmp_seq=14 ttl=64 time=1.00 ms
64 bytes from 192.168.64.5: icmp_seq=15 ttl=64 time=1.70 ms
64 bytes from 192.168.64.5: icmp_seq=16 ttl=64 time=1.95 ms
64 bytes from 192.168.64.5: icmp_seq=17 ttl=64 time=2.18 ms
64 bytes from 192.168.64.5: icmp_seq=18 ttl=64 time=1.89 ms
^Z
zsh: suspended ping 192.168.64.5

kali@kali:~$ nmap -sS 192.168.64.5
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-30 19:49 -0600
Nmap scan report for 192.168.64.5
Host is up (0.00062s latency).
All 1000 scanned ports on 192.168.64.5 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: EA:B8:2C:4E:F2:51 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 0.66 seconds

kali@kali:~$
```

nmap after:



3.2 Vulnerability Scanning (Nikto)

A web vulnerability scan was conducted using Nikto against the Apache server.

Command used:

```
nikto -h http://192.168.64.6
```

Key findings include:

- Outdated Apache version
- Missing security headers:
 - Content-Security-Policy
 - Strict-Transport-Security
 - X-Content-Type-Options
 - Referrer-Policy
- Potential information disclosure via ETags

These issues indicate misconfigurations that could be exploited by attackers.

Screenshot(s):

```
Session Actions Edit View Help
kali@kali:~$ nmap -sS 192.168.64.5
Starting Nmap 7.90 ( https://nmap.org ) at 2026-03-30 20:47 -0800
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.52 seconds

kali@kali:~$ nmap -sS 192.168.64.6
Starting Nmap 7.90 ( https://nmap.org ) at 2026-03-30 20:47 -0800
Nmap scan report for 192.168.64.6
Host is up (0.00070s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
MAC Address: EA:18872C:4E:F2:151 (Unknown)
Nmap done: 1 IP address (1 host up) scanned in 0.68 seconds

kali@kali:~$ niko -u http://192.168.64.6
- Nikto v2.6.0

+ Your Nikto installation is out of date.
+ Target IP:      192.168.64.6
+ Target Hostname: 192.168.64.6
+ Target Port:    80
+ Platform:       Linux/Unix
+ Start Time:     2026-03-30 20:49:46 (GMT-6)

+ Server: Apache/2.4.58 (Ubuntu)
+ No CGI Directories found (use '-C all' to force check all possible dirs). C
GI tests skipped.
+ [999984] /: Server may leak inodes via ETags, header found with file /, ino
de: 29af, size: 64e48eaf5310b, mtine: gzip. See: https://cve.mitre.org/cgi-bi
n/cvename.cgi?name=CVE-2003-1418
+ [013587] /: Suggested security header missing: strict-transport-security. S
ee: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Strict-Transpor
t-Security
+ [013587] /: Suggested security header missing: content-security-policy. See
: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Permissions-Policy
+ [013587] /: Suggested security header missing: x-content-type-options. See:
https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Content-Type-Opt
ions
+ [013587] /: Suggested security header missing: permissions-policy. See: htt
ps://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Permissions-Policy
+ [013587] /: Suggested security header missing: referer-policy. See: https:
//developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Referer-Policy
+ [600050] Apache/2.4.58 appears to be outdated (current is at least 2.4.66).
+ [999990] OPTIONS: Allowed HTTP Methods: HEAD, GET, POST, OPTIONS .
+ [007342] /: X-Frame-Options header is deprecated and was replaced with the Content-Security-Policy HTTP header with the frame-ancestors directive. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Reference/Headers/X-Frame-Options
+ [007352] /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabilit
ies/missing-content-type-header/
+ 813 requests, 16 errors and 10 items reported on the remote host
+ End Time:      2026-03-30 20:55:14 (GMT-6) (328 seconds)
+ 1 host(s) tested

kali@kali:~$
```

4. Network Traffic Analysis (Wireshark)

Network traffic was captured using Wireshark during the Nmap scan to analyze attack patterns.

Observations:

- Multiple TCP SYN packets were sent from the attacker (192.168.64.4) to the target (192.168.64.6), indicating a port scanning attempt.
- Open ports responded with **SYN-ACK** packets.
- The attacker terminated connections using **RST packets** instead of completing the TCP handshake.
- Closed ports responded with **RST-ACK packets**.

This pattern is characteristic of a **TCP SYN scan**, which is commonly used for stealthy reconnaissance.

Screenshot(s):

Wireshark capture showing a list of network packets. The filter is `ip.addr == 192.168.64.6`. The packet list shows various TCP connections and resets. The packet details pane shows the structure of a packet, including Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. The packet bytes pane shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
35	72.589872839	192.168.64.4	192.168.64.6	TCP	58	60821 → 8080 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
36	72.589932214	192.168.64.4	192.168.64.6	TCP	58	60821 → 110 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
37	72.589941589	192.168.64.4	192.168.64.6	TCP	58	60821 → 1723 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
38	72.589947881	192.168.64.4	192.168.64.6	TCP	58	60821 → 199 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
39	72.589954756	192.168.64.4	192.168.64.6	TCP	58	60821 → 8888 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
40	72.589960881	192.168.64.4	192.168.64.6	TCP	58	60821 → 143 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
41	72.589966298	192.168.64.4	192.168.64.6	TCP	58	60821 → 111 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
42	72.589973048	192.168.64.4	192.168.64.6	TCP	58	60821 → 1025 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
43	72.589979089	192.168.64.4	192.168.64.6	TCP	58	60821 → 80 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
44	72.589984173	192.168.64.4	192.168.64.6	TCP	58	60821 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
45	72.591853255	192.168.64.6	192.168.64.4	TCP	54	8080 → 60821 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
46	72.591853672	192.168.64.6	192.168.64.4	TCP	54	110 → 60821 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
47	72.591853765	192.168.64.6	192.168.64.4	TCP	54	1723 → 60821 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
48	72.591853787	192.168.64.6	192.168.64.4	TCP	54	199 → 60821 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
49	72.591853838	192.168.64.6	192.168.64.4	TCP	54	8888 → 60821 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
50	72.591853880	192.168.64.6	192.168.64.4	TCP	54	143 → 60821 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
51	72.591853963	192.168.64.6	192.168.64.4	TCP	54	111 → 60821 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
52	72.591854065	192.168.64.6	192.168.64.4	TCP	54	1025 → 60821 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
53	72.592249047	192.168.64.6	192.168.64.4	TCP	58	80 → 60821 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
54	72.592249297	192.168.64.6	192.168.64.4	TCP	54	23 → 60821 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
55	72.592283172	192.168.64.4	192.168.64.6	TCP	54	60821 → 80 [RST] Seq=1 Win=0 Len=0
56	72.592345963	192.168.64.4	192.168.64.6	TCP	58	60821 → 21 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
57	72.592351547	192.168.64.4	192.168.64.6	TCP	58	60821 → 113 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
58	72.592355088	192.168.64.4	192.168.64.6	TCP	58	60821 → 554 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
59	72.592359546	192.168.64.4	192.168.64.6	TCP	58	60821 → 1720 [SYN] Seq=0 Win=1024 Len=0 MSS=1460

Wireshark capture showing a list of network packets. The filter is `tcp.flags.syn == 1`. The packet list shows various TCP connections and resets. The packet details pane shows the structure of a packet, including Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. The packet bytes pane shows the raw data in hexadecimal and ASCII.

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38	72.589947881	192.168.64.4	192.168.64.6	TCP	58	60821 → 199 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
39	72.589954756	192.168.64.4	192.168.64.6	TCP	58	60821 → 8888 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
40	72.589960881	192.168.64.4	192.168.64.6	TCP	58	60821 → 143 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
41	72.589966298	192.168.64.4	192.168.64.6	TCP	58	60821 → 111 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
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43	72.589979089	192.168.64.4	192.168.64.6	TCP	58	60821 → 80 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
44	72.589984173	192.168.64.4	192.168.64.6	TCP	58	60821 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
53	72.592249047	192.168.64.6	192.168.64.4	TCP	58	80 → 60821 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
56	72.592345963	192.168.64.4	192.168.64.6	TCP	58	60821 → 21 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
57	72.592351547	192.168.64.4	192.168.64.6	TCP	58	60821 → 113 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
58	72.592355088	192.168.64.4	192.168.64.6	TCP	58	60821 → 554 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
59	72.592359546	192.168.64.4	192.168.64.6	TCP	58	60821 → 1720 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
60	72.592363588	192.168.64.4	192.168.64.6	TCP	58	60821 → 22 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
61	72.592367838	192.168.64.4	192.168.64.6	TCP	58	60821 → 25 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
62	72.592371995	192.168.64.4	192.168.64.6	TCP	58	60821 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
63	72.592375646	192.168.64.4	192.168.64.6	TCP	58	60821 → 3389 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
64	72.592379213	192.168.64.4	192.168.64.6	TCP	58	60821 → 135 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
65	72.592382796	192.168.64.4	192.168.64.6	TCP	58	60821 → 445 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
66	72.592385963	192.168.64.4	192.168.64.6	TCP	58	60821 → 587 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
67	72.592402880	192.168.64.4	192.168.64.6	TCP	58	60821 → 3308 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
68	72.592411796	192.168.64.4	192.168.64.6	TCP	58	60821 → 993 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
69	72.592415213	192.168.64.4	192.168.64.6	TCP	58	60821 → 256 [SYN] Seq=0 Win=1024 Len=0 MSS=1460

5. Analysis

The results demonstrate how attackers identify services and probe systems for weaknesses. The presence of missing security headers and outdated software increases the attack surface of the web server.

The traffic analysis highlights how such attacks can be detected by monitoring TCP flags and identifying abnormal connection patterns such as repeated SYN packets.

6. Mitigation

To reduce the identified risks, the following measures are recommended:

- Keep Apache and system packages updated
 - Configure proper HTTP security headers (e.g., CSP, HSTS)
 - Use firewalls to restrict unnecessary open ports
 - Monitor network traffic for suspicious patterns such as SYN scans
-

7. Conclusion

This project demonstrates the complete lifecycle of a basic cyber attack, from reconnaissance to detection. By combining tools such as Nmap, Nikto, and Wireshark, it is possible to identify vulnerabilities and detect malicious activity in a controlled environment.

This lab highlights the importance of both offensive and defensive security techniques in modern cybersecurity.